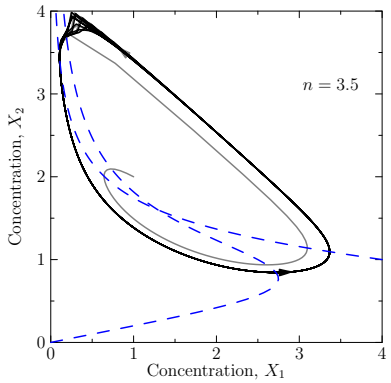
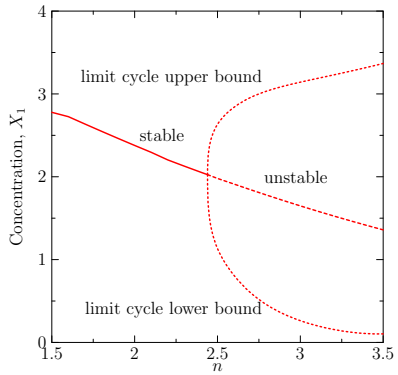
Computational Biology

Lecture 18: *Dynamical Systems*

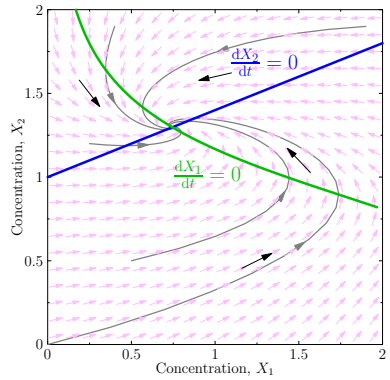




ODEs, nullclines, fixed points, stability analysis, bifications

Outline

1. **Understanding ODEs**
2. Stability
3. Bifications



Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X})$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Two Species

- Consider the simple system

$$\varnothing \xrightarrow{f(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$
$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

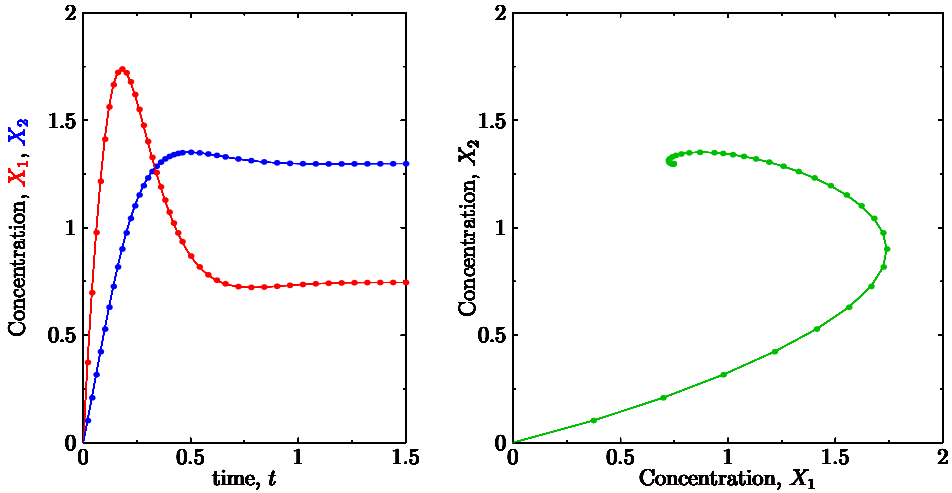
- Where X_2 represses the expression of X_1

$$f(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$

- Leading to the coupled ODEs

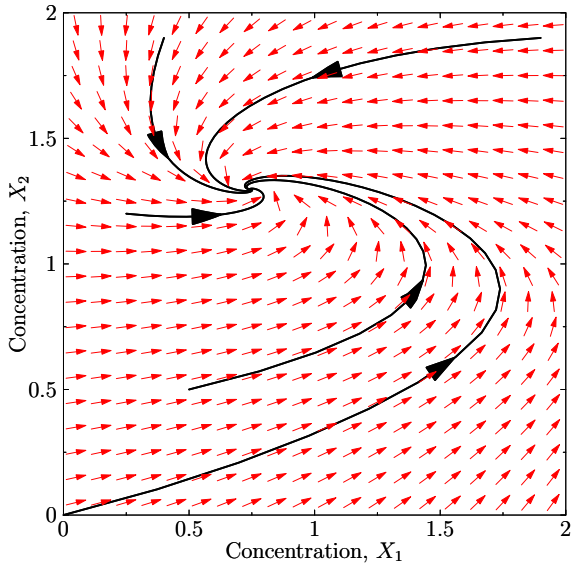
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Trajectories



Nullclines

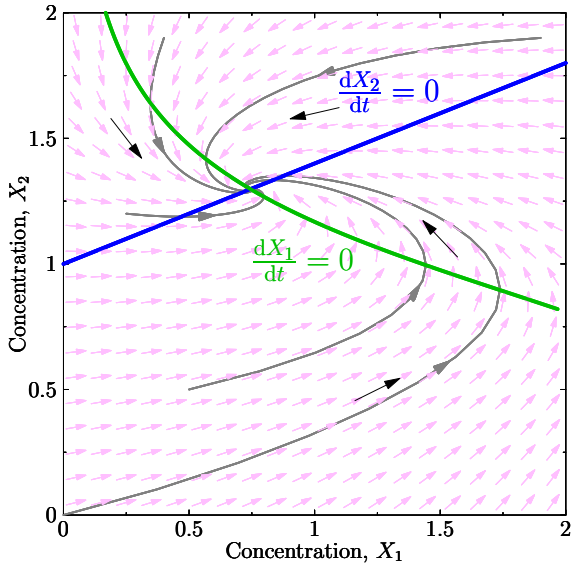
- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$
- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1+(X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \quad \Rightarrow\ X_1 = \frac{k_1}{(k_3+k_5)(1+(X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \qquad \Rightarrow\ X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

- The intersection of the nullclines defines the stationary points

Nullclines



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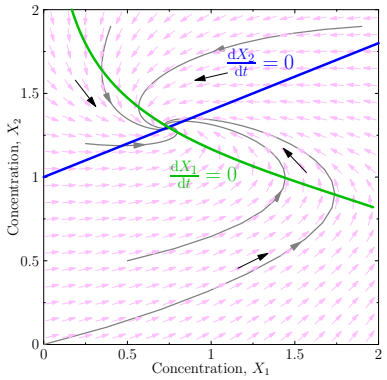


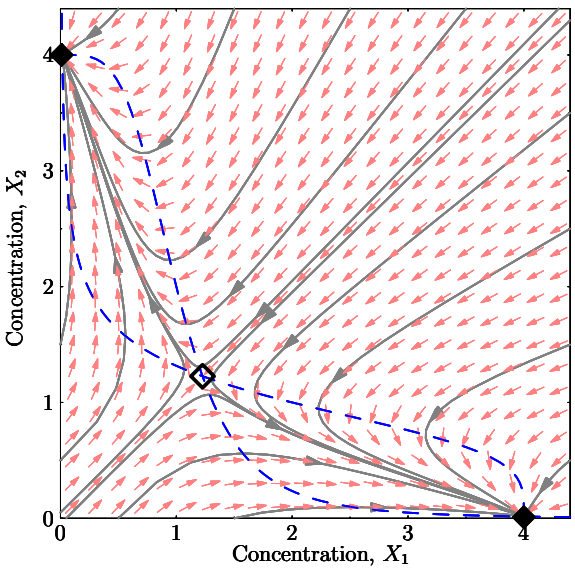
Bistability

- Let's consider a slightly different system where X_1 and X_2 inhibits each others' expression

$$\frac{dX_1(t)}{dt} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{dX_2(t)}{dt} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

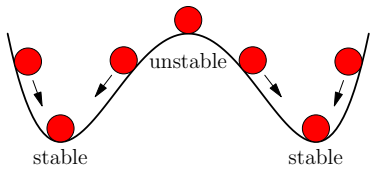
- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ we end up with three fixed points

Nullclines and Bistablity

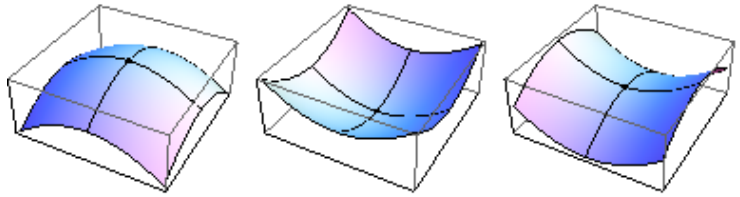


Stability

- Points can be stable or unstable



- Above 1D we can also have saddle-points



Jacobian

- To study stability at a fixed point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

- or

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where $\mathbf{J}$ is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

Linear Approximation

- Let $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$

- To solve this we note that we can write most square matrices $\mathbf{J}$ as

$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Where $\boldsymbol{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} \approx \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

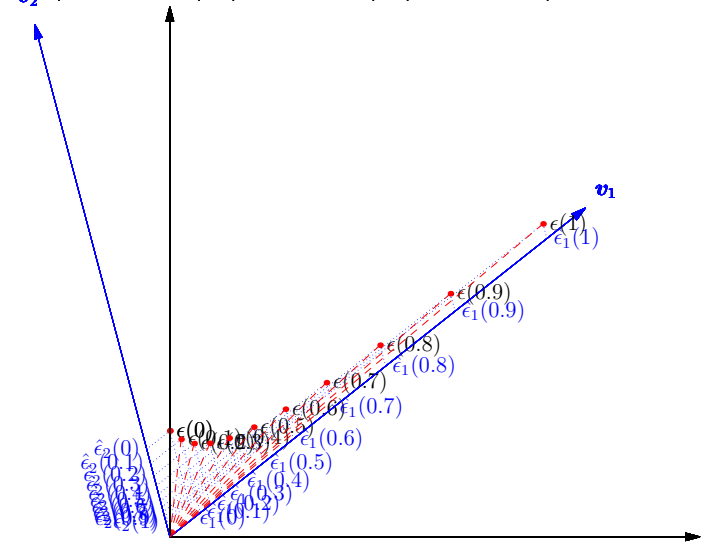
$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} \approx \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} \approx \lambda_i\,\hat{\epsilon}_i(t)$$

- With solution

$$\hat{\epsilon}_i(t) \approx \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Oscillator

- Consider a chain with a positive feedback

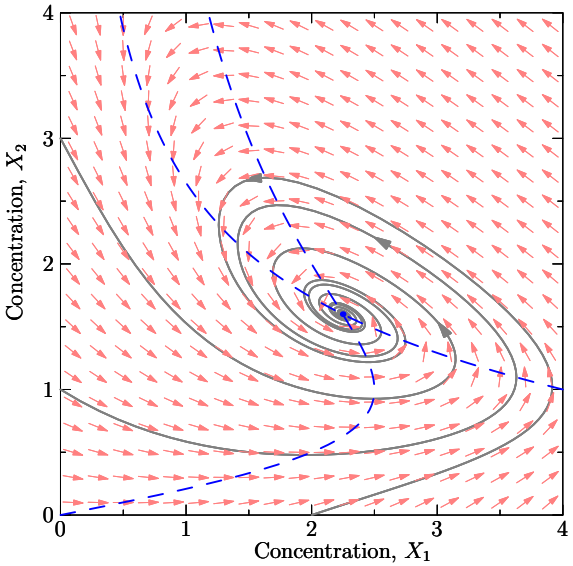
$$\varnothing \xrightarrow{k_0} X_1 \qquad f(X_2) = k_1\,(1 + (X_2/K)^n)$$

$$X_1 \xrightarrow{f(X_2)} X_2 \qquad \frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$X_2 \xrightarrow{k_2} \varnothing \qquad \frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\,X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

Nullclines and Bistablity



Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\,X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$
- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1\left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K}\left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1\left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K}\left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

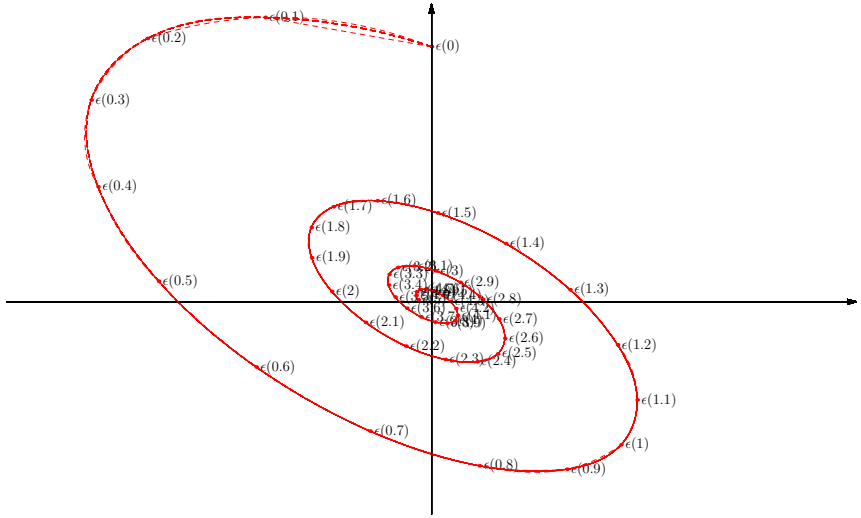
$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11}\, J_{22} - J_{12}\, J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



High Dimensional Systems

- We have only considered simple systems with two species of molecules

- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions

- Using programs like Mathematica it is often easy to find fixed point solutions numerically

- However, in biology different reactions happen at very different rates

- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

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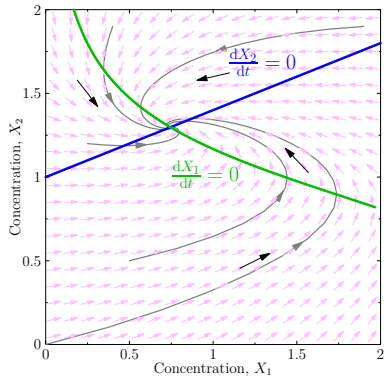


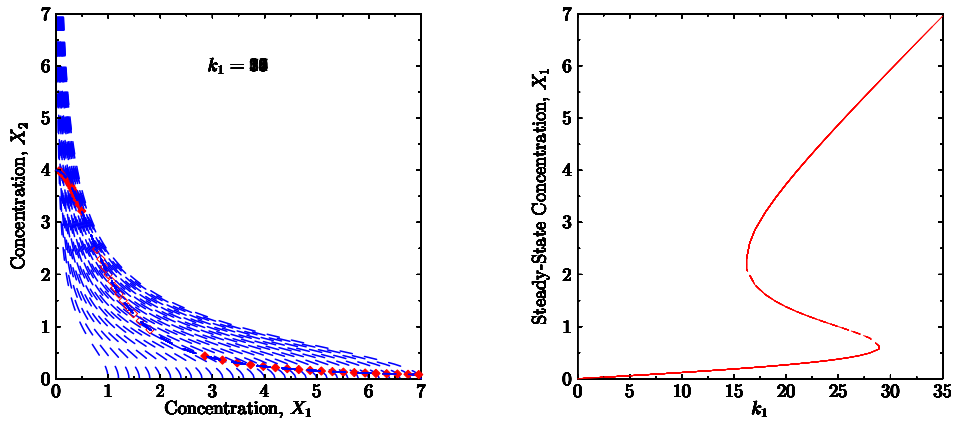
Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

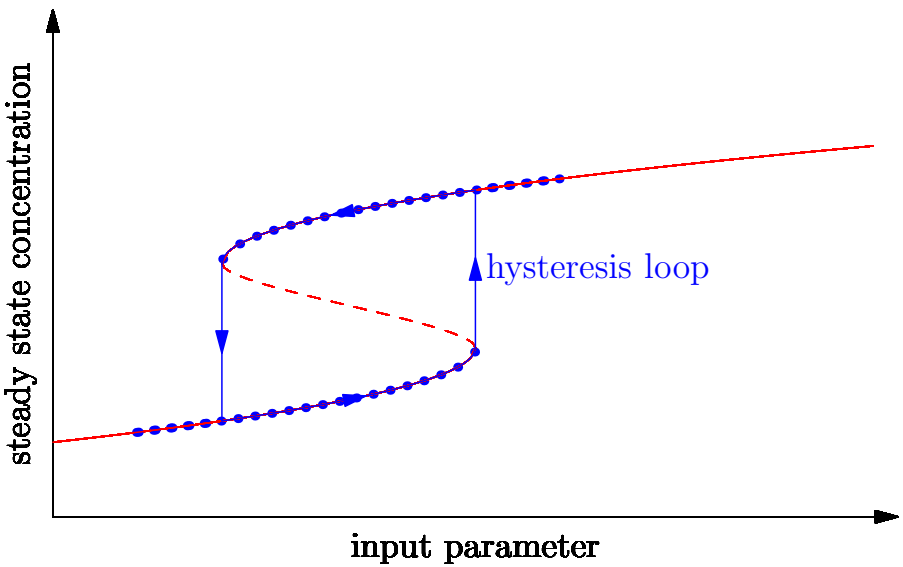
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- Let $k_2 = 20,\ K_1 = K_2 = 1,\ k_3 = k_4 = 5,\ n_1 = n_2 = 2$
- But now we look at varying the parameter k_1
- The fixed points are given where the nullclines intersect

Varying k_1



Hysteresis



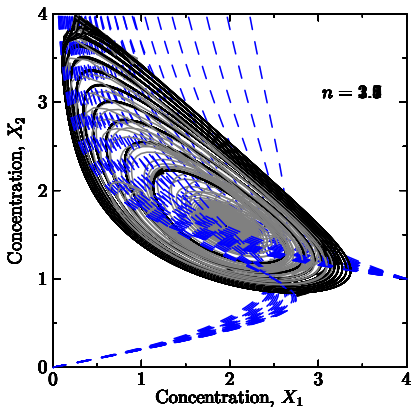
Autocatalysis

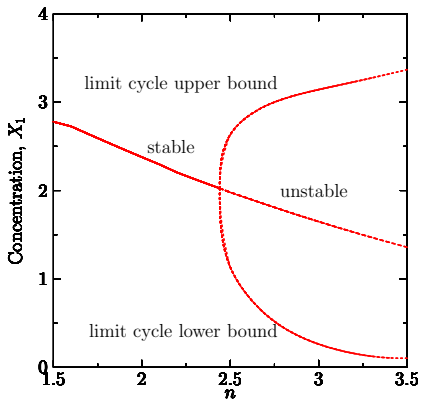
- Recall the autocatalytic system

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8,\ k_1 = 1,\ K = 1,\ k_2 = 5$
- We consider what happens when we vary n

Hopf Bifurcation





Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable